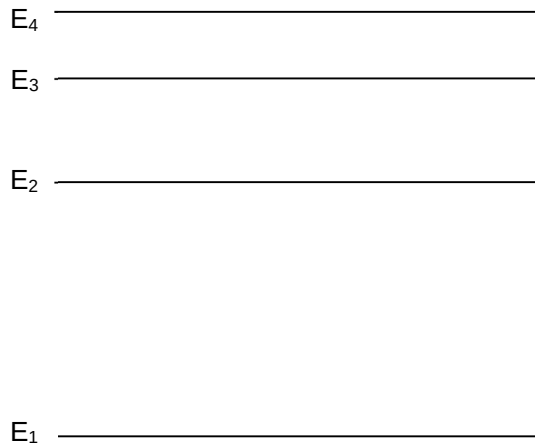


26 The diagram represents in simplified form some of the energy levels of the hydrogen atom.



The transition of an electron from E_3 to E_2 is associated with the emission of red light.

Which transition could be associated with the emission of blue light?

A E_4 to E_1

B E_1 to E_4

C E_4 to E_2

D E_2 to E_4

Ans: C

$\lambda_{red} \approx 700 \text{ nm}$, $\lambda_{blue} \approx 400 \text{ nm}$, $E = \frac{hc}{\lambda}$ hence the energy transition for blue

light should be about 1.75 x that of red light.

Transition is from a higher energy level to lower energy level for emission of light.